Solution to Quiz

1. Test the series for convergence or divergence

   (a)
   $$\sum_{n=1}^{\infty} \frac{n-1}{2n+1}$$

   (b)
   $$\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$$

   (c)
   $$\sum_{n=1}^{\infty} ne^{-n^2}$$

   (d)
   $$\sum_{n=1}^{\infty} \frac{1}{2+3^n}$$

   (e)
   $$\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$$

   Solution:
   (a)
   $$\lim_{n\to\infty} \frac{n-1}{2n+1} = \frac{1}{2}$$
   diverges by the Divergence Test.

   (b) **Test:** We can apply the **Limit Comparison Test** by comparing it with the series $\sum_{n=1}^{\infty} \frac{n^{3/2}}{n^3} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$.

   In fact, for large n, the numerator behaves like $\sqrt{n^3} = n^{3/2}$ and the denominator behaves like $3n^3$. Hence, the series behaves like:

   $$\frac{n^{3/2}}{3n^3} = \frac{1}{3n^{3/2}}$$

   This is a **p-series** with $p = \frac{3}{2}$, and since $p > 1$, this series **converges**.

   Limit Comparison Test with $b_n = \frac{1}{n^{3/2}}$

   Analysis: $\lim_{n\to\infty} \frac{\sqrt{n^3+1}/(3n^3)}{1/n^{3/2}} = \frac{1}{3}$

   Conclusion: $\boxed{\text{Converges}}$

   (c) We can apply the Limit Comparison Test. Note that e^{-n^2} decays extremely fast as $n \to \infty$, much faster than any polynomial. The terms go to zero rapidly as $n \to \infty$.

   To confirm, we can compare it with the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$, which is a convergent p-series.

Limit Comparison Test with $b_n = \frac{1}{n^2}$

Analysis: $\lim_{n\to\infty} \frac{ne^{-n^2}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^3}{e^{n^2}} = \lim_{x\to\infty} \frac{x^3}{e^{x^2}} \overset{\text{H}}{=} 0$

Conclusion: Converges

(d) **Series:** $\sum_{n=1}^{\infty} \frac{1}{2+3^n}$

Test: Direct Comparison Test with $b_n = \frac{1}{3^n}$

Analysis: $\frac{1}{2+3^n} < \frac{1}{3^n}$ for all $n \geq 1$

Conclusion: Converges

(e)We can apply the Integral Test to determine the convergence of this series. We consider the integral of the function $f(x) = \frac{1}{x(\ln(x))^2}$ from $x = 2$ to infinity.

We can use a substitution $u = \ln(x)$, which leads to:

$$\int_2^{\infty} \frac{1}{x(\ln(x))^2} dx = \int_{\ln 2}^{\infty} \frac{du}{u^2} = \frac{1}{\ln 2}$$

Since the integral converges, the series converges by the Integral Test.

2. Find the sum of the series

$$\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right)$$

Solution: geometric series shows that

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{k\to\infty} \sum_{n=1}^{k} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \lim_{k\to\infty} \left(1 - \frac{1}{k+1} \right) = 1$$

So the final answer is $3 \times 1 + 1 = 4$.

3. Consider the following series:

$$\sum_{n=1}^{\infty} \left(-\frac{1}{n} \right)^n.$$

(a) Show that the series is convergent;

(b) How many terms of the series do we need to add in order that the partial sum S_n is within 0.00005 of the actual sum?

Solution:

(a) The series $\sum_{n=1}^{\infty} \left(-\frac{1}{n} \right)^n = \sum_{n=1}^{\infty} (-1)^n \frac{1}{n^n}$ satisfies (i) of the Alternating Series Test because $\frac{1}{(n+1)^{n+1}} < \frac{1}{n^n}$ and (ii) $\lim_{n\to\infty} \frac{1}{n^n} = 0$, so the series is convergent.

(b) Now $b_5 = \frac{1}{5^5} = 0.00032 > 0.00005$ and $b_6 = \frac{1}{6^6} \approx 0.00002 < 0.00005$, so by the Alternating Series Estimation Theorem, $n = 5$. (That is, since the 6 th term is less than the desired error, we need to add the first 5 terms to get the sum to the desired accuracy.)

4. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

$$(a)\ \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$$

$$(b)\ \sum_{n=1}^{\infty} \left(\frac{1-n}{2+3n}\right)^n$$

$$(c)\ \sum_{n=1}^{\infty} \frac{(-9)^n}{n10^{n+1}}$$

$$(d)\ \sum_{n=1}^{\infty} \frac{\sin(n\pi/6)}{1+n\sqrt{n}}$$

Solution:

(a) $\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$ converges by the Alternating Series Test since $\lim_{n\to\infty} \frac{1}{\ln n} = 0$ and $\left\{\frac{1}{\ln n}\right\}$ is decreasing. Now $\ln n < n$, so $\frac{1}{\ln n} > \frac{1}{n}$, and since $\sum_{n=2}^{\infty} \frac{1}{n}$ is the divergent (partial) harmonic series, $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ diverges by the Comparison Test. Thus, $\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$ is conditionally convergent.

(b) $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{1-n}{2+3n}\right)^n\right|} = \lim_{n\to\infty} \frac{n-1}{3n+2} = \lim_{n\to\infty} \frac{1-1/n}{3+2/n} = \frac{1}{3} < 1$, so the series $\sum_{n=1}^{\infty} \left(\frac{1-n}{2+3n}\right)^n$ is absolutely convergent by the Root Test.

(c) $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(-9)^{n+1}}{(n+1)10^{n+2}} \cdot \frac{n10^{n+1}}{(-9)^n}\right| = \lim_{n\to\infty} \left|\frac{(-9)n}{10(n+1)}\right| = \frac{9}{10} \lim_{n\to\infty} \frac{1}{1+1/n} = \frac{9}{10}(1) = \frac{9}{10} < 1$, so the series $\sum_{n=1}^{\infty} \frac{(-9)^n}{n10^{n+1}}$ is absolutely convergent by the Ratio Test.

(d) Since $|\sin(n\pi/6)| \le 1$ for all n, we have:

$$\left|\frac{\sin(n\pi/6)}{1+n\sqrt{n}}\right| \le \frac{1}{1+n\sqrt{n}} < \frac{1}{n^{3/2}}$$

As the series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ is a p-series with $p = \frac{3}{2} > 1$, which is known to converge. By the comparison test, the series $\sum_{n=1}^{\infty} \left|\frac{\sin(n\pi/6)}{1+n\sqrt{n}}\right|$ converges, implying that the original series $\sum_{n=1}^{\infty} \frac{\sin(n\pi/6)}{1+n\sqrt{n}}$ is absolutely convergent.

5. Determine the radius and interval of convergence of the following series:

(a)

$$\sum_{k=0}^{\infty} \frac{k}{4^k} x^k$$

(b)

$$\sum_{k=1}^{\infty} \frac{1}{k}(x-1)^{2k+1}$$

(c)

$$\sum_{k=2}^{\infty} \frac{(k!)^2}{(2k)!} x^{2k+1}$$

Solution:

(a) Using the Ratio Test,

$$\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty} \left|\frac{(k+1)4^k}{k4^{k+1}}\right|$$
$$= \frac{1}{4} \lim_{k\to\infty} \frac{k+1}{k} = \frac{1}{4}$$

So the radius of convergence is 4 and the series converges absolutely for $x \in (-4, 4)$. When $x = 4$,

$$\sum_{k=0}^{\infty} \frac{k}{4^k} 4^k = \sum_{k=0}^{\infty} k \text{ and } \lim_{k\to\infty} k = \infty,$$

so the series diverges by the Divergence Test. When $x = -4$,

$$\sum_{k=0}^{\infty} \frac{k}{4^k} (-4)^k = \sum_{k=0}^{\infty} (-1)^k k$$

which diverges by the Divergence Test.

Thus the interval of convergence is $(-4, 4)$ and $r = 4$.

(b) Write $a_k = \frac{1}{k}(x-1)^{2k+1}$. Using the Ratio test

$$\begin{aligned}
\lim_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right| &= \lim_{k\to\infty} \left| \frac{(x-1)^{2k+3} k}{(x-1)^{2k+1}(k+1)} \right| \\
&= (x-1)^2 \lim_{k\to\infty} \frac{k}{k+1} \\
&= (x-1)^2
\end{aligned}$$

Note that

$$\begin{aligned}
&(x-1)^2 < 1 \\
&\text{when} \\
&-1 < x - 1 < 1 \\
&\implies 0 < x < 2
\end{aligned}$$

So, the series converges absolutely for $x \in (0, 2)$. When $x = 0, \sum_{k=1}^{\infty} \frac{(-1)^{2k+1}}{k} = -\sum_{k=1}^{\infty} \frac{1}{k}$ is the divergent harmonic series; when $x = 2, \sum_{k=1}^{\infty} \frac{1}{k}$ is the divergent harmonic series. So, the interval of convergence is $(0, 2)$ and $r = 1$.

(c) Write $a_k = \frac{(k!)^2}{(2k)!} x^{2k+1}$. Using the Ratio Test, we have

$$\begin{aligned}
\lim_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right| &= \lim_{k\to\infty} \left| \frac{(k+1)^2}{(2k+2)(2k+1)} x^2 \right| \\
&= \left| \frac{1}{4} x^2 \right| < 1
\end{aligned}$$

for $-2 < x < 2$. So, the series converges for $-2 < x < 2$ and diverges for $x < -2$ or $x > 2$. Since the Ratio Test gives no conclusion at $x = -2$ and $x = 2$, we must test separately at $x = -2$ and $x = 2$.

$$\begin{aligned}
a_k &= \frac{(k!)^2}{(2k)!} x^{2k+1} \\
&= \frac{(k!)^2}{(2k)!} 2^{2k+1} \\
&= \frac{(k!)}{1.3 \cdot 5 \cdot 6 \ldots . (2k-1)} 2^{k+1} \\
&= \frac{(2 \cdot 4 \cdot 6 \cdot 8 \ldots . (2k))}{1 \cdot 3 \cdot 5 \cdot 6 \ldots (2k-1)} \cdot 2 > 4
\end{aligned}$$

Since $\lim_{k\to\infty} a_k \neq 0$, the given series diverges for $x = 2$. Similarly the given series diverges at $x = -2$. Thus the power series converges at all x in the interval $(-2, 2)$ and diverges at all x outside of this interval. The radius of convergence is $r = 2$.

6. Use a known Taylor series to find the Taylor series about $a = 0$ for the given function and find its radius of convergence.

(a) $f(x) = \frac{e^x - 1}{x}$

(b) $f(x) = x \sin 2x$

Solution:

(a)

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

with $r = \infty$. We see that

$$\begin{aligned} \frac{e^x - 1}{x} &= \frac{\sum_{k=0}^{\infty} x^k/k! - 1}{x} \\ &= \sum_{k=1}^{\infty} \frac{x^{k-1}}{k!} = \sum_{k=0}^{\infty} \frac{x^k}{(k+1)!} \end{aligned}$$

with $r = \infty$.

(b)

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

with $r = \infty$. Replacing x with $2x$ and multiplying by x, we have

$$\begin{aligned} x \sin 2x &= x \left(\sum_{k=0}^{\infty} \frac{(-1)^k (2x)^{2k+1}}{(2k+1)!} \right) \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k+1} x^{2k+2}}{(2k+1)!} \end{aligned}$$

with $r = \infty$.